# Homework 5

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**6.1 Consider again the example application of Bayes rule in Section 6.2.1. Suppose the doctor decides to order a second laboratory test for the same patient, and suppose the second test returns a positive result as well. What are the posterior probabilities of cancer and ¬cancer following these two tests? Assume that the two tests are independent.**

**Answers:** Assuming that the two sets are independent between each other.

$$P(+,+\mid cancer) = P(+\mid cancer)P(+\mid cancer)$$

then the posterior probabilities of cancer is

$$P(cancer\mid +,+) = \frac{P(+,+\mid cancer)P(cancer)}{P(+,+)} = \frac{P(+\mid cancer)P(+\mid cancer)P(cancer)}{P(+,+)}$$

The posterior probabilities of ¬cancer is

$$P(\neg cancer\mid +,+) = \frac{P(+,+\mid \neg cancer)P(\neg cancer)}{P(+,+)} = \frac{P(+\mid \neg cancer)P(+\mid \neg cancer)P(\neg cancer)}{P(+,+)}$$

P(+|cancer) P(+|cancer) P(cancer)=0.98*0.98*0.008=0.0076832

P(+|¬cancer) P(+|¬cancer) P(¬cancer)=0.03*0.03*0.992=0.0008928

P(+,+) = P(+,+|cancer) P(cancer) + P(+,+|¬cancer)P(¬cancer)

=0.0076832+0.0008928=0.008576

Therefore, P(cancer|+,+)＝0.0076832/0.008576=0.895896

P(¬cancer|+,+)=0.104104

**6.2 In the example of Section 6.2.1 we computed the posterior probability of cancer by normalizing the quantities P (+|cancer) • P (cancer) and P (+|¬cancer) • P (¬cancer) so that they summed to one, Use Bayes theorem and the theorem of total probability (see Table 6.1) to prove that this method is valid (i.e., that normalizing in this way yields the correct value for P(cancer|+)).**

**Answers:** Based on formulas, we can know $P(cancer\mid +) = \frac{P(+\mid cancer)P(cancer)}{P(+)}$ .

Because cancer event and ¬cancer event are mutually exclusive, and

$$P(cancer) + P(\neg cancer) = 1 .$$

From total probability formula,

$$P(+) = P(+\mid cancer)P(cancer) + P(+\mid \neg cancer)P(\neg cancer)$$

We have

$$\begin{aligned} P(cancer\mid +) &= \frac{P(+\mid cancer)P(cancer)}{P(+)} \\ &= \frac{P(+\mid cancer)P(cancer)}{P(+\mid cancer)P(cancer) + P(+\mid \neg cancer)P(\neg cancer)} \end{aligned}$$

Then this method is proved to be valid.

**6.3 Consider the concept learning algorithm FindG, which outputs a maximally general consistent hypothesis (e.g., some maximally general member of the version space).**
**(a) Give a distribution for P(h) and P(D|h) under which FindG is guaranteed to output a MAP hypothesis.**
**(b) Give a distribution for P(h) and P(D|h) under which FindG is not guaranteed to output a MAP hypothesis.**
**(c) Give a distribution for P(h) and P(D|h) under which FindG is guaranteed to output a ML hypothesis but not a MAP hypothesis.**
**Answers:**
(a)P(h): if hypothesis $h_1$ is more general than $h_2$, make $P(h_1)>=P(h_2)$ .

$$P(D|h)=\begin{cases}1 & \forall d_i, d_i = h(x_i)\\ 0 & otherwise\end{cases}$$

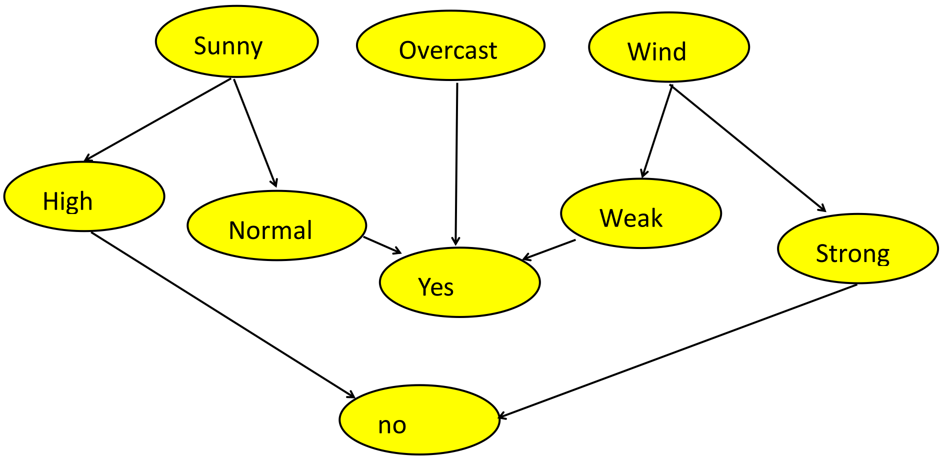
(b)P(h): if hypothesis $h_1$ is more general than $h_2$, make $P(h_1)<=P(h_2)$ .

$$P(D|h)=\begin{cases}1 & \forall d_i, d_i = h(x_i)\\ 0 & otherwise\end{cases}$$

(c)P(h): for every hypothesis $h_i$ and $h_j$, make $P(h_i) = P(h_j) = \frac{1}{|H|}$

$$P(D|h)=\begin{cases}1 & \forall d_i, d_i = h(x_i)\\ 0 & otherwise\end{cases}$$

**6.6 Draw the Bayesian belief network that represents the conditional independence assumptions of the naive Bayes classifier for the PlayTennis problem of Section 6.9.1. Give the conditional probability table associated with the node Wind.**
**Answers:**




| | Strong | Weak |
|---|---|---|
| yes | 0.33 | 0.67 |
| no | 0.60 | 0.40 |